



A stochastic simulation-based optimization method for equitable and efficient network-wide signal timing under uncertainties



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ABSTRACT

The equity of right-of-way is an important topic in traffic management and control. With the balance consideration of traffic equity and efficiency, which are respectively evaluated by the Atkinson index and average travel time, this study proposes a bi-objective signal timing simulation-based optimization (SO) model under uncertainties, and solve it by a bi-objective stochastic simulation-based optimization (BOSSO) method. In this method, two types of surrogate models (i.e., regressing Kriging model and quadratic regression model) are used to capture the complicated mapping relationship between decision variables and bi-objectives, respectively in the whole variable domain and in the local trust-region. Meanwhile, the incorporation of the global regressing Kriging model and an adaptive selector helps to predict bi-objective values of untested samples and re-estimate simulated samples in the local trust-region, which can save great computational costs and smooth stochastic noises. Moreover, the non-interactive role of a decision maker is taken to generate more Pareto optimal solutions around his/her desired bi-objective values. Through the algorithm comparison for a benchmark bi-objective stochastic optimization problem, the proposed BOSSO method is validated to outperform three other counterparts (i.e., NSGA-II, BOTR and BOEGO) under the same simulation costs. In real-field experiments, an urban road network with 15 signalized and five non-signalized intersections in Changsha, China is modeled by VISSIM. After the well calibration of the microscopic traffic simulation model, the network-wide bi-objective signal timing stochastic SO problems with and without coordination are solved by BOSSO. Numerical results indicate that compared with the real-field case, the average travel time and Atkinson index are reduced respectively by at most 13.48% and 23.49% for optimized non-coordinated signal plans, and respectively by at most 25.58% and 2.83% for optimized coordinated ones. It is further validated that under variable traffic volumes, the non-coordinated signal plan can well improve both traffic efficiency and equity, and the coordinated one is capable to improve traffic efficiency at a larger degree but sacrifice traffic equity. Moreover, the balance analyses show the existence of competing relationship between bi-objectives, and BOSSO is confirmed to outperform NSGA-II, BOTR and BOEGO in searching the better Pareto optimal signal plans under the same budgeted simulations. In conclusion, BOSSO is promising to address bi-objective

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optimization problems characterized by costly evaluation, high dimensions and stochastic noises.

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1. Introduction

Previous studies on the traffic signal control optimization considering the equity issue are nowhere near those traffic efficiency-oriented studies. However, the freeway ramp closure experiment for eight weeks implemented by Department of Transportation, Minnesota opens an important chapter in traffic control design concerning the equity issue (Levinson et al., 2002). This experiment validates that the ramp control benefits long distance travelers by sacrificing the benefits of short distance travelers, which implies its ineffectiveness (inequity) to short distance travelers. Hence, to gain the public acceptance and support for this ramp control system, it is necessary to consider not only the improvement of systematic traffic efficiency, but also how such improvement is distributed across various groups of travelers. In this condition, when designing the traffic control system, the traffic efficiency is treated as an evaluation indicator rather than the only indicator. Moreover, Kotsialos and Papageorgiou (2004), Zhang and Levinson (2004) and Kesten et al. (2013) found that the higher efficient control system exhibits the worse traffic equity, which implies the existence of the competing relationship between traffic efficiency and equity. While, Yin et al. (2004) proposed a fairer ramp control algorithm, which meanwhile maintains a similar level of overall traffic efficiency. Zhang and Shen (2010) revealed that traffic efficiency and equity can be simultaneously improved, and there may not be a conflicting relationship between each other. Therefore, it is necessary to re-visit the equitable and efficient traffic control problem so as to address above contradictory conclusions.

Regarding the application of equity index in traffic control optimizations, Levinson and Zhang (2006) investigated the temporal and spatial inequities of ramp metering control. The former evaluates the differences of travel time (or delay/speed) for drivers on the same route, who reach the on-ramp at different times. The latter evaluates the differences of travel time (or delay/speed) for drivers, who arrive at various on-ramps at the same time. Litman (2007) proposed the lateral and vertical equity evaluation methods. The former does not consider the personal attributes, while the latter takes into account the distribution of gains or losses among individuals or groups with various travel demands and personal attributes. Kesten et al. (2013) employed various statistical models or socio-economic models to judge the equity of traffic control strategies, which mainly include relative mean difference, logarithmic variance, Gini coefficient, Theil index, Atkinson index, Kolm index. Notably, Gini coefficient (Gini, 1936) and its related Lorenz curve (Lorenz, 1905) that describe the social income gap are popular in capturing the equity of traffic control system (Levinson et al., 2002; Yin et al., 2004; Levinson and Zhang, 2004). However, Atkinson (1970) pointed out that the shape of the Lorenz curve may be infinitely varied without any change in the Gini coefficient. In addition, Atkinson (1970) and Cowell (2011) also stated that for typical distributions more weight would be attached by Gini coefficient to transfers in the centre of the distribution than at the tails, and it is not clear that such a weighting would necessarily accord with social values.

Moreover, Papageorgiou and Kotsialos (2001) integrated the delay and the weighted penalty term of the equity constraint about on-ramps as the objective function of a ramp control optimization model. Ma and Zhang (2008) combined the total travel time, aggregated and non-aggregated equity indices into a weighted objective, and proposed a bi-criteria corridor control design problem with Cellular Transmission Model (CTM) as the evaluation tool. Then, a heuristic algorithm was employed to solve its optimal green split and ramp control rate. Khoo (2011) presented a nonlinear ramp control programming model with consideration of traffic efficiency and the equity constraint, and solved it by a dynamic penalty approach. Meng and Khoo (2010) proposed a spatial equity index to describe the distribution of delay across various on-ramps, and built a dynamic ramp control optimization model with the modified CTM as the evaluation tool. Its Pareto optimal solutions about the total delay and equity index were solved by non-dominated sorting genetic algorithm II (NSGA-II) (Deb et al., 2002). Tian et al. (2012) proposed other equity indices and ramp control strategies based on travel demand and space, and found that the conflict between the traffic efficiency and equity gradually vanishes as the travel demand elasticity decreases.

However, to our best knowledge, very limited literatures involve the urban traffic signal timing optimization with equity considerations. For example, Li and Ge (2014) proposed a multi-objective bi-level programming model to design the sustainable and equitable signal timing for a congested urban traffic network. Baskan and Ozan (2017) also proposed a bi-level programming model to address the urban traffic signal timing problem by considering the reserve capacity with an equity constraint. Unfortunately, they are all analytical mathematical models, which may not well capture the sophisticated relationships between the signal timing plan and network-wide traffic states. Nowadays, the mature microscopic traffic simulation software and secondary development technologies provide an alternative way to flexibly design and simulate the complicated interactions between various elements of urban road traffic system. However, taking the microscopic traffic simulation as the evaluation tool has the following drawbacks. (1) The deviation in objective values caused by many stochastic processes in the traffic simulation can lead to the failure of traditional optimization methods designed for noise-free optimization problems. (2) Due to the high computational cost and running time, it is difficult to be used together with optimization algorithms that rely on extensive exploration of entire domain, e.g., Genetic Algorithm (GA), Tabu search, etc. (3) “Black-box” property of the traffic simulation makes it difficult to derive the explicit expression that fits classic

optimization formulations (e.g., gradient descent). Fortunately, many simulation-based optimization (SO) methods were proposed in past decades to address these weaknesses. For more details, please refer to [Andradóttir S \(1998\)](#), [Fu \(1994, 2002\)](#), [Fu et al. \(2005\)](#), [Barton and Meckesheimer \(2006\)](#), [Hachicha et al. \(2010\)](#), and [Pasupathy and Ghosh \(2013\)](#).

Therefore, this study will put forward a bi-objective stochastic simulation-based optimization (BOSSO) method to solve a network-wide signal timing optimization problem with balance consideration of traffic equity and efficiency under uncertainties. The remainder of this paper is organized as follows. [Section 2](#) reviews the related works about SO methods. [Section 3](#) formulates the stochastic simulation-based bi-objective signal timing optimization problem and proposes the BOSSO method. [Section 4](#) studies a benchmark problem and the real-field experiments. Finally, some important findings are concluded in [Section 5](#).

2. Literature reviews about SO methods

SO approaches address the optimization problems whose objective functions (or constraints) are evaluated by computationally expensive simulations. Here, only surrogate model (meta-model) based SO techniques are specifically reviewed considering the focus of this study. Their critical step is to construct an analytical approximation of simulation-based components of the optimization problem (e.g., objective function, constraints) with the limited information from evaluated samples. According to [Søndergaard \(2003\)](#), the surrogate models can be classified into physical and functional types. In detail, the formulation and parameters of physical surrogate models have a physical or structural interpretation, and provide the analytically tractable and global information, which are application- or problem- specific. On the other hand, the functional surrogate models can approximate locally any kind of objective functions (or constraints) and do not take into account any information on specific underlying problems. They are chosen based on their analytical tractability, which include the low-order polynomials ([Kleijnen, 2008](#); [Conn et al., 2009](#); [Ryu and Kim, 2014](#)), spline models ([Barton and Meckesheimer, 2006](#)), radial basis functions (RBFs) ([Regis and Shoemaker, 2005](#); [Wild et al., 2008](#); [Ouvray and Bierlaire, 2009](#); [Zhou et al., 2013](#)), Kriging models ([Booker et al., 1999](#); [Kleijnen et al., 2010](#); [Chen et al., 2014, 2016](#); [He et al., 2017](#)), and support vector machines ([Chen et al., 2014](#)). Another type of combined surrogate model proposed by [Osorio and Bierlaire \(2013\)](#) takes full advantages of both physical and functional surrogate models. However, it is usually difficult to derive a physical surrogate model (e.g., an analytical and differentiable macroscopic traffic model) under a complex scenario. Meanwhile, the physical surrogate model would be less detailed and less accurate than the evaluation outputs from the simulator. Thus, it is popular to use the functional surrogate model instead of the physical one (or the combined one) to approximate the objective function or constraints. In the traffic optimization domain, there are usually three types of SO methods to address the multi-objective SO problems.

- (a) Multi-objectives are formulated into a weighted objective, which converts the multi-objective SO problem into a single-objective SO problem. Then, a SO method is utilized to search the global optima, e.g., the derivative-free trust region method ([Osorio and Nanduri, 2015a, b](#)) and the evolutionary algorithm ([Khondaker and Kattan, 2015](#)). However, these studies lack the balance analyses between various objectives.
- (b) Some multi-objective evolutionary algorithms (EAs) are applicable to address the multi-objective SO problems. For example, the multiple-objective genetic algorithm (MOGA) was employed to solve the traffic signal control SO problems with balance consideration of traffic efficiency and safety ([Stevanovic et al., 2013](#)) when employing the VISSIM simulation as the evaluation tool, or total delay and vehicular emissions ([Zhang et al., 2013](#)) with the CTM as the evaluation tool. [Meng and Khoo \(2010\)](#) utilized NSGA-II to address the dynamic ramp metering optimization problem with concerns of total delay and equity index. Besides, [Jin \(2011\)](#) reviewed the surrogate model based EAs that can be applied in the single- and multi-objective SO problems.
- (c) One surrogate model is constructed for each objective function with costly simulation samples, and then intuitive infill strategies are designed to search new solutions to dominate at least one current Pareto optimal solution. In this way, the Pareto front is iteratively updated and improved. For instance, [He et al. \(2017\)](#) employed two regressing Kriging models to approximate two objective functions of the time-varying pricing optimization problem, and then used the strategies of Probability of Improvement (PI) and Expected Improvement (EI) to update the Pareto front. This method is referred as bi-objective efficient global optimization (BOEGO) here. Similarly, other surrogate model based multi-objective optimization algorithms in [Ryu and Kim \(2014\)](#), [Kim and Ryu \(2011\)](#), and [Müller \(2017\)](#) can be also transplanted to solve similar SO problems.

To address the type of multi-objective SO problem with high-dimensional variables and stochastic noises, this study attempts to propose the BOSSO method and apply it to search the reliable signal timing plans with the balance consideration of traffic efficiency and equity.

3. Methodology

The urban transportation system is dynamic, stochastic, non-linear and complex. It is not sufficient for a simplified and analytical mathematical model to formulate the sophisticated interactions between various elements of the traffic system, e.g., vehicle and vehicle, signal control plan and network-wide traffic states. However, the mature microscopic traffic simulation software and secondary development technologies provide us an alternative way. For example, the popular microscopic

traffic simulator VISSIM applied here offers a friendly Graphical User Interface (GUI), which can easily design any type of road networks. Meanwhile, it also provides the mature COM interface to flexibly modify the interactions between various traffic elements, and allow the inter-process communication and data exchange between VISSIM and other compatible software (e.g., Matlab, Python). Hence, this type of microscopic traffic simulator is qualified to capture the complicated mapping relationship between the signal plan and network-wide traffic states, and its output high-resolution vehicular travel data also benefit the accurate evaluation of traffic equity and efficiency. This section will firstly present two evaluation indices for traffic equity and efficiency respectively, which are based on the individual vehicular travel times from VISSIM simulation. Then, the network-wide bi-objective signal timing SO problem is formulated under uncertainties. After that, the BOSSO method is proposed to solve this type of optimization problem.

3.1. Bi-objective signal timing stochastic Simulation-based optimization model

Equity index: Atkinson index proposed by Atkinson (1970) is to evaluate the inequity of social incomes or wealth. Inspired by this measure of inequity, it is assumed that each traveler achieves some benefits (i.e., “income” or “wealth”) once he/she reaches his/her destination through the signalized road network. Generally, the lower travel time means the higher benefit the traveler obtains. In order to compare the benefits achieved by travelers from the same and various routes, the travel time ratio of vehicle i is introduced as

$$t_{r,i} = t_i / \bar{t}_{j_i} \quad (1)$$

where t_i is the travel time of individual vehicle i , j_i denotes the route that vehicle i traverses, and $\bar{t}_{j_i}$ is the average travel time of route j_i , c.f., Eq. (2).

$$\bar{t}_{j_i} = \sum_{i=1}^{|j_i|} t_i / |j_i| \quad (2)$$

where $|j_i|$ indicates the number of vehicles from route j_i . That is also to say, the average travel time of route j_i are measured by the average travel time of all vehicles from route j_i .

Then, Atkinson index (A) based on travel time ratios is formulated to evaluate the traffic equity, c.f., Eq. (3).

$$A(t_{r,1}, t_{r,2}, \dots, t_{r,N_v}) = \begin{cases} 1 - \frac{1}{\bar{\mu}} \left(\frac{1}{N_v} \sum_{i=1}^{N_v} t_{r,i}^{1-\varepsilon} \right)^{1/(1-\varepsilon)} & \text{for } 0 \leq \varepsilon \neq 1 \\ 1 - \frac{1}{\bar{\mu}} \left(\prod_{i=1}^{N_v} t_{r,i} \right)^{1/N_v} & \text{for } \varepsilon = 1 \end{cases} \quad (3)$$

where N_v is the total number of vehicles, $\bar{\mu}$ is the mean travel time ratio, the Atkinson is often called the “inequality aversion parameter”. For $\varepsilon = 0$ (no aversion to inequity), it is assumed that no social benefit is gained by complete redistribution of individual travel times, and $A = 0$. For $\varepsilon = \infty$ (infinite aversion to inequity), it is assumed that infinite social benefit is obtained by complete redistribution of individual travel times, and $A = 1$. Thus, A varies between 0 and 1. Given a ε value, the lower value of A represents the more equal distribution of travel time ratios than the higher value.

In this condition, the minimization of A can be analyzed as follows. As for vehicles from the same route, their travel times tend to be the same and then to be equal to the average travel time of that route. As for vehicles from different routes, their travel time ratios tend to be the same. So, it is perfectly equitable (i.e., $A = 0$) if vehicles from the same route have the equal travel time and vehicles from different routes have the same travel time ratio. To describe the relative change of traffic equity under various signal plans, the relative Atkinson index is given as

$$A_r = A/A_0 \quad (4)$$

where A_0 is the Atkinson index evaluated by the calibrated traffic simulator with the benchmark signal plan.

Efficiency index: Based on individual travel times from the high-resolution microscopic traffic simulator, the average travel time is employed to evaluate the traffic efficiency. Similarly, to reflect the relative change of traffic efficiency under various signal plans, the relative index of average travel time is formulated as

$$T_r = \frac{(\sum_{i=1}^{N_v} t_i) / N_v}{\bar{T}_0} \quad (5)$$

where $\bar{T}_0$ is the average travel time evaluated by the calibrated traffic simulator with the benchmark signal plan.

Moreover, due to stochastic noises involved in the microscopic traffic simulation (e.g., randomness of driving behaviors), one input of signal timing plan x will result in various values of A_r and T_r output from the simulation. Then, the network-wide bi-objective signal timing stochastic SO problem is formulated as

$$\begin{aligned} \text{Minimize } f(x, \zeta) &= \{E[A_r(x, \zeta)], E[T_r(x, \zeta)]\} \\ \text{s.t. } \Phi(x, \zeta) &\rightarrow \{A_r(x, \zeta), T_r(x, \zeta)\} \end{aligned} \quad (6)$$

where $\Omega_1 = \{x \in \mathbb{R}^n: \mathbf{a}_1 \leq x \leq \mathbf{b}_1\}$ is the search space of decision variables, $\mathbf{a}_1$ and $\mathbf{b}_1$ respectively denote the lower and upper bounds of decision vector x with the dimension n . ζ denotes the exogenous parameters in the traffic simulator,

which cover the fixed parameters (e.g., road network structure, and signal phase setting) and stochastic parameters (e.g., randomness of driving behaviors and routing behaviors). $\Phi(x, \varsigma)$ represents the microscopic traffic simulation with the inputs of x and ς . $\rightarrow$ stands for the output operator. $A_r(x, \varsigma) = a_r(x, \varsigma) + \delta_a, \delta_a \sim N(0, \sigma_a^2)$ and $T_r(x, \varsigma) = t_r(x, \varsigma) + \delta_t, \delta_t \sim N(0, \sigma_t^2)$. That is, $A_r(x, \varsigma)$ and $T_r(x, \varsigma)$ indicate the **sampled bi-objective values with stochastic noises**, respectively, and $a_r(x, \varsigma)$ and $t_r(x, \varsigma)$ are **true expected values of bi-objectives**. Two random variables δ_a and δ_t are assumed to follow normal distribution. $E[\cdot]$ is the expectation operator. In this study, x is composed of green timings of all signal phases for the non-coordinated case, or consists of green timings and offsets for the coordinated case. Due to the discrete integer type of x , it becomes more difficult to solve the problem (6) as n gets larger.

Obviously, the problem (6) aims to simultaneously minimize two expectations of A_r and T_r . Usually, average approximation method (AAM) is a popular approach (Fitzpatrick and Grefenstette, 1988) to address stochastic noises δ_a and δ_t in the formulation ((6)). That is, the sampled bi-objective values by some repetitive simulations with a given x and ς are averaged to denote the expectations of two objectives. A larger number of repetitions will smooth the noises better and make the average values closer to the true expectations, which however greatly increase the computational costs. Thus, AAM is not an ideal choice to deal with the noises for the stochastic SO problem. Fortunately, some other classic methods provide useful inspirations, such as, the informational approach to global optimization (IAGO) (Villemonteix et al., 2009), the RBF-based surrogate model for global optimization (Jakobsson et al., 2010), the sequential Kriging optimization (SKO) (Huang et al., 2006), the regressing Kriging model-based EI algorithm (Chen et al., 2014; He et al., 2017). The following solution method will take full advantages of the regressing Kriging model to address the stochastic noises.

3.2. Bi-objective stochastic Simulation-based optimization method

To solve the box-constrained bi-objective stochastic SO problem (6), a bi-objective derivative-free trust region (BOTR) algorithm proposed by Ryu and Kim (2014) deserves our attentions. Three reasons are given as follows. (I) BOTR is a type of derivative-free algorithm, which is just applicable to the black-box optimization problem (e.g., SO problem); (II) It can produce the well-spread non-dominated solutions with limited computational costs, unlike NSGA-II that needs thousands of function evaluations and BOEGO that searches only one improved solution for each iteration; (III) It has been proved to have the well convergence performance. However, BOTR is not qualified to address the bi-objective stochastic SO problem with the high dimensions. The explanations are stated as follows. When the decision vector is high-dimensional, a large number of samples in the local trust-region are required to be evaluated to build the critical quadratic regression models between decision variables and bi-objectives. This would cause unaffordable computational costs if the computationally expensive simulation is taken as the evaluation tool. Meanwhile, BOTR never involves how to deal with noises during bi-objective function evaluations.

Thus, this section proposes the BOSSO method based on BOTR. Three critical points of BOSSO are stated as follows. (1) Two regressing Kriging models built with global simulated samples are used to predict (re-estimate) bi-objective values of unevaluated (simulated) samples in the local trust-region. (2) An adaptive selector is introduced to determine some of the poised set of samples in the local trust-region to be evaluated by simulation. (3) A decision maker's non-interactive role is incorporated to impact the search direction and trust-region by his/her desired bi-objective values. The first critical point can tolerate stochastic noises from simulation outputs and smooth their negative effect in the following steps and iterations. The incorporation of the first two points can save a lot computational costs when fitting the high-dimensional quadratic regression models with satisfactory accuracies. The third point benefits to guiding one part of non-dominated solutions toward the desired bi-objective values, which completely depend on the decision maker's preference. In this condition, above three critical points make the BOSSO method be capable to address the bi-objective stochastic SO problem with high dimensions and non-interactive role of a decision maker. The detailed steps of BOSSO are presented in the following.

Initialization: Set the initial point $x^{(0)}$ with the dimension n , and evaluate the bi-objective function values $F(x^{(0)}) = [f_1(x^{(0)}), f_2(x^{(0)})]$. The initial set of non-dominated points is $\chi^{(0)} = \{x^{(0)}\}$. The initial trust-region radius is $\Delta^{(0)}(x^{(0)}) \in (0, \Delta_{\max}]$. Set the maximum and tolerable trust-region radiuses as $\Delta_{\max} > 0$ and $\Delta_{\text{tol}} \geq 0$, respectively. The minimum closeness allowed between bi-objective values is $\delta > 0$. The critical reduction ratio is $\eta \in [0, 1)$. The reduction coefficient about the trust-region for the inner iteration is set as $\omega \in (0, 1)$. The decrease and increase parameters about the trust-region for the outer iteration are set as $\tau_{\text{dec}} \in (0, 1)$ and $\tau_{\text{inc}} > 1$, respectively. Two constant coefficients about the norm of the gradient of quadratic regression model are $\mu > \beta > 0$. The global simulated and filtered sample sets are denoted as $Z = \emptyset$ and $X = \emptyset$, respectively.

Algorithm: For each **outer iteration** $k = 1, 2, \dots$,

Step 1 (Iteration Determination): Return the set of non-dominated points $\chi^{(k-1)}$ from the previous iteration. Select the most isolated point $x_c^{(k)}$ with the trust region radius $\Delta^{(k-1)}(x_c^{(k)})$ from $\chi^{(k-1)}$ (c.f., Appendix A). Then, set the reference point $r^{(k)} = F(x_c^{(k)})$, $Y^{(k)} = \emptyset$, and $\Delta_0^{(k)} = \Delta^{(k-1)}(x_c^{(k)})$.

Step 2 For each **inner iteration** $j = 1, 2, \dots$,

Step (i) (Determine the current number of samples): Set $\tilde{\Delta}^{(k)} = \omega^{j-1} \cdot \Delta_0^{(k)}$, generate a poised set of samples $S_p = \{y_1^{(j)}, y_2^{(j)}, \dots, y_d^{(j)}\}$ in $B(x_c^{(k)}, \tilde{\Delta}^{(k)})$ by Latin Hypercube Sampling (LHS). Use an adaptive selector to select d_c samples

$S_c = \{y_1^{(j)}, y_2^{(j)}, \dots, y_{d_c}^{(j)}\}$ randomly from S_p (c.f., Eq. (7)), which will be evaluated by the costly simulation.

$$d_c = \min \left\{ d_{\max}, \max \left(d_{\min}, \left\lceil \log(k) / \left(c_f \cdot (\tilde{\Delta}^{(k)})^2 \right) \right\rceil \right) \right\} \quad (7)$$

where $d_{\min}$ and $d_{\max}$ respectively denote the minimum and maximum number of selected samples, c_f is a positive constant, and $d_{\min} \leq d_c \leq d_{\max} \leq d$. Intuitively, when the radius $\tilde{\Delta}^{(k)}$ is large and the solutions are distant from the theoretical Pareto front, the highly accurate quadratic regression models (c.f., Eq. (9)) are not needed. Then, d_c is relatively small. As k increases, the radius $\tilde{\Delta}^{(k)}$ decreases and the solutions become closer to the theoretical Pareto front. Then, a larger number of simulated samples are required to fit more accurate quadratic regression models, which contributes to producing the more frontier and widely spread ultimate non-dominated solutions.

Step (ii) (Build the global regressing Kriging model): Evaluate the selected d_c samples by costly simulation, and add these evaluated samples to Z : $Z \leftarrow Z \cup S_c$. Employ Z to build two regressing Kriging models (c.f., Appendix B).

Step (iii) (Prediction and re-estimation): Apply the built regressing Kriging models to predict the unevaluated samples in S_p (i.e., $S_p \setminus S_c$), and re-estimate samples in S_c to filter noises from the simulation evaluations (c.f., Appendix B). Add the filtered samples to $Y^{(k)}$ and X : $Y^{(k)} \leftarrow Y^{(k)} \cup S_c$, and $X \leftarrow X \cup S_c$.

Step (iv) (Quadratic regression model): Based on the poised sample set S_p and their predicted and filtered bi-objectives f_1 and f_2 , construct the quadratic regression model $\tilde{m}_i^{(j)}$ of $f_i^{(k)}$: $\tilde{m}_i^{(j)}(x_c^{(k)} + s) = \tilde{c}_i^{(j)} + s^T \tilde{g}_i^{(j)} + \frac{1}{2} s^T \tilde{H}_i^{(j)} s$, $i = 1, 2, 3$, where $\tilde{g}_i^{(j)}$ and $\tilde{H}_i^{(j)}$ are the gradient and Hessian of $\tilde{m}_i^{(j)}$ at $s = 0$. Notably, the third single-objective function f_3 is combined by f_1 and f_2 , which is formulated as

$$f_3^{(k)}(x) = f_3(x; r^{(k)}) = - \prod_{i=1}^2 \left\{ (r_i^{(k)} - f_i^{(k)}(x))_+ \right\}^2 \quad (8)$$

where $f_i^{(k)} = f_i$, and $r^{(k)} = (r_1^{(k)}, r_2^{(k)}) = (f_1^{(k)}(x_c^{(k)}), f_2^{(k)}(x_c^{(k)}))$. Specifically, $m_i(x)$ is formulated as

$$m_i(x) = \alpha_{i,0} + \sum_{k=1}^n \alpha_{i,k} x_k + \sum_{k=1}^n \alpha_{i,k+n} x_k^2 \quad (9)$$

where $\alpha_{i,0}$, $\alpha_{i,k}$ and $\alpha_{i,k+n}$ are the parameters to be estimated for f_i and x_k is the k^{th} component of x .

Step (v) (Criticality checking step): If $\tilde{\Delta}^{(k)} \leq \max\{\Delta_{\text{tol}}, \min_{i=1,2,3} \mu \|\tilde{g}_i^{(j)}\|\}$, then set $m_i^{(k)} = \tilde{m}_i^{(j)}$ and $\Delta_c^{(k)} = \min[\max\{\tilde{\Delta}^{(k)}, \beta \min_{i=1,2,3} \|\tilde{g}_i^{(j)}\|\}, \Delta_0^{(k)}]$, and go to Step 3. Otherwise, increase j by one and go to Step (i).

Step 3 (Three single-objective problems): Solve each single-objective problem in the trust-region $B(x_c^{(k)}, \Delta_c^{(k)})$ and let $z_i^{(k)} = \arg \min\{m_i^{(k)}(x) : x \in B(x_c^{(k)}, \Delta_c^{(k)})\}$. The minimizer of $m_3^{(k)}$ (for $f_3^{(k)}$) is a non-dominated solution in $B(x_c^{(k)}, \Delta_c^{(k)})$, and those of $m_1^{(k)}$ and $m_2^{(k)}$ (respectively for $f_1^{(k)}$ and $f_2^{(k)}$) are employed to expand the solution set toward both ends of the theoretical Pareto front. Then, evaluate $z_i^{(k)}$ ($i = 1, 2, 3$) by the costly simulation, and add these three evaluated samples to Z : $Z \leftarrow Z \cup \{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$. Re-build these two regressing Kriging models with Z , and use the updated regressing Kriging models to re-estimate three samples $\{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$ to smooth the stochastic noises from the simulation evaluations. Add the three filtered solutions to X : $X \leftarrow X \cup \{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$.

Step 4 (Reduction ratios): Compute the reduction ratio for each newly evaluated point as follows.

For $y \in Y^{(k)}$, its reduction ratio is $\rho^{(k)}(y) = \frac{f_3^{(k)}(x_c^{(k)}) - f_3^{(k)}(y)}{m_3^{(k)}(x_c^{(k)}) - m_3^{(k)}(z_3^{(k)})}$;

For $z \in \{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$, let $\rho^{(k)}(z) = \frac{f_i^{(k)}(x_c^{(k)}) - f_i^{(k)}(z)}{m_i^{(k)}(x_c^{(k)}) - m_i^{(k)}(z)}$, $i = 1, 2, 3$;

For $x \in \chi^{(k-1)}$, set $\rho^{(k)}(x) = \rho^{(k-1)}(x)$.

Remove sample points with insufficient reduction ratio from $Y^{(k)}$: $Y^{(k)} \leftarrow \{y \in Y^{(k)} : \rho^{(k)}(y) \geq \eta\}$.

Step 5 (Trust-region radius update): Design a trust-region radius for each newly evaluated point.

For $y \in Y^{(k)}$, set $\Delta^{(k)}(y) \in [\Delta_c^{(k)}, \min\{\tau_{\text{inc}} \cdot \Delta_c^{(k)}, \Delta_{\text{max}}\}]$;

For $z \in \{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$, set $\Delta^{(k)}(z) \in \begin{cases} [\Delta_c^{(k)}, \min\{\tau_{\text{inc}} \Delta_c^{(k)}, \Delta_{\text{max}}\}] & \text{if } \rho^{(k)}(z) \geq \eta \\ \{\tau_{\text{dec}} \Delta_c^{(k)}\} & \text{otherwise.} \end{cases}$;

For $x \in \chi^{(k-1)}$, set $\Delta^{(k)}(x) \in \begin{cases} \Delta_c^{(k)} & \text{if } x = x_c^{(k)} \\ \Delta^{(k-1)}(x) & \text{otherwise.} \end{cases}$.

Step 6 (Non-dominated solution set update): Compare vector values $F(x) = [f_1(x), f_2(x)]$ for $x \in \chi^{(k-1)} \cup Y^{(k)} \cup \{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$, and give the updated set of non-dominated points $\chi^{(k)}$.

Step 7 (**Iteration evaluation**): Examination of successful iteration and termination.

- (a) **Successful iteration**: If $\max \{\Delta^{(k)}(x) : x \in \chi^{(k)}\} \geq \Delta_{tol}$ and there exist $x \in \chi^{(k)} \setminus \chi^{(k-1)}$ such that $\rho^{(k)}(x) \geq \eta$, then increase k by one, and go to Step 1.
- (b) **Unsuccessful iteration**: Else if $\max \{\Delta^{(k)}(x) : x \in \chi^{(k)}\} \geq \Delta_{tol}$, for all $x \in \chi^{(k)} \setminus \chi^{(k-1)}$, $\rho^{(k)}(x) < \eta$, and $\tau_{dec} \cdot \Delta_c^{(k)} > \Delta_{tol}$, then set $x_c^{(k+1)} = x_c^{(k)}$, $\Delta_0^{(k+1)} = \tau_{dec} \cdot \Delta_c^{(k)}$, increase k by one, and go to Step 2.
- (c) **Termination**: Otherwise, terminate the algorithm.

Output Pareto front and optimal solutions for Z and those for X.

4. Numerical experiments

To validate the effectiveness and superiority of BOSSO, this section will firstly apply BOSSO to solve a benchmark problem called FON function (Fonseca and Fleming, 1998) with stochastic parameters. Then, BOSSO is employed to calibrate the microscopic traffic simulation model for a real-field study site, and solve the network-wide bi-objective signal timing stochastic SO problems with and without coordination. In these numerical experiments, BOSSO is also compared with other three counterparts (i.e., NSGA-II, BOTR and BOEGO) in terms of the convergence performance, diversity of solutions and computational costs. Finally, under variable traffic volumes, bi-objective values for three cases are compared from overall and local traffic efficiency and equity.

Before carrying out the numerical experiments, two indices to judge the wellness of non-dominated solutions are introduced as follows. A well-known measure named Generational Distance (GD) (Veldhuizen, 1999) is used to evaluate the closeness of the generated solutions to the theoretical Pareto front, and a Hypervolume (HV) indicator to judge the diversity of these solutions (Ryu and Kim, 2014). The former is calculated by

$$GD = \sqrt{\sum_{j=1}^v \|F(x_j) - F(x_j^*)\|^2 / v} \quad (10)$$

where v denotes the number of solutions (i.e., $S = \{x_1, x_2, \dots, x_v\}$) generated by the algorithm. $F(x_j^*)$ ($j = 1, 2, \dots, v$) indicates the Pareto efficient point that is the closest to $F(x_j)$, and x_j^* is the Pareto optimal solution corresponding to $F(x_j^*)$. If all solutions in S are Pareto optimal, GD becomes zero.

However, in real-field applications, the Pareto efficient points (i.e., theoretical Pareto front) cannot be known in advance or even never be known. Thus, it is necessary to design a meta-Pareto front H (with a meta-Pareto solution set Ω') so as to search the Pareto efficient points. That is, the meta-Pareto front is set as ultimate non-dominated points by one algorithm or those by various algorithms (c.f., Figs. 5(a) and 6(a)). Then, the revised GD is formulated as

$$GD^* = \sqrt{\sum_{j=1}^v \|F(x_j) - H(x_j^*)\|^2 / v} \quad (11)$$

If $x_j \in \Omega'$, then $x_j^* = x_j$ and $H(x_j^*) = F(x_j)$; else, $H(x_j^*)$ is the meta-Pareto efficient point that is the closest to $F(x_j)$.

Besides, the value of HV is equal to the area enclosed by the generated solutions and a reference point (i.e., $R_H = [f_1(x_0), f_2(x_0)]$) that must be dominated by all $F(x_j)$, $j = 1, 2, \dots, v$. Thus, the larger HV is the better the diversity will be. With the solution set S sorted in the increasing order concerning f_1 , HV is computed by

$$HV = \sum_{i=0, \dots, v-1} \{f_1(x_0) - f_1(x_{i+1})\} \{f_2(x_i) - f_2(x_{i+1})\} \quad (12)$$

4.1. Algorithm comparison for a benchmark problem

This section aims to validate the better performance of BOSSO in dealing with stochastic noises and searching well-spread non-dominated solutions than other three algorithms (i.e., NSGA-II, BOTR and BOEGO). Here, the FON function is taken as the benchmark bi-objective optimization problem, which has a theoretical non-convex Pareto front (c.f., Fig. 1). Meanwhile, some noises are added to the parameters about the decision vector x so as to generate the stochastic FON function, c.f., Eq. (13).

$$\begin{aligned} \text{Minimize : } g_1(x, \xi) &= 1 - \exp\left(-\sum_{i=1}^3 \left(x_i \cdot \xi_i - \frac{1}{\sqrt{3}}\right)^2\right) \\ g_2(x, \xi) &= 1 - \exp\left(-\sum_{i=1}^3 \left(x_i \cdot \xi_i + \frac{1}{\sqrt{3}}\right)^2\right) \end{aligned} \quad (13)$$

where the noise $\xi_i \sim N(1, 0.1)$ and the decision variable $x_i \in [-2, 2]$. Its theoretical Pareto-optimal solutions are $x_1 = x_2 = x_3 \in [-1/\sqrt{3}, 1/\sqrt{3}]$. To address the stochastic noises, NSGA-II and BOTR both apply AAM, i.e., twice function evaluations are

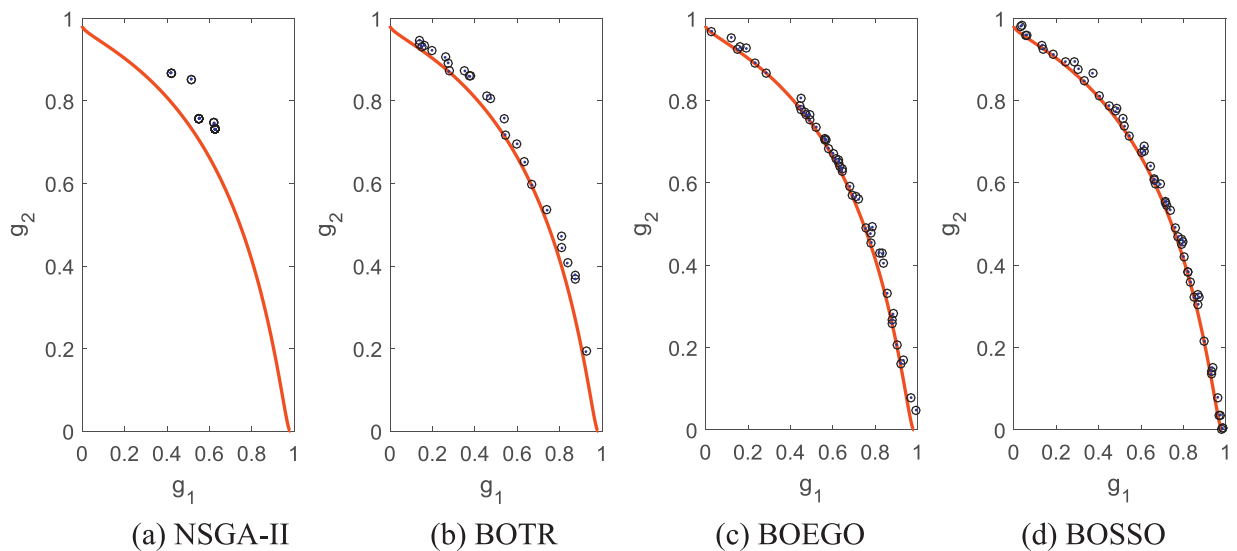


Fig. 1. Comparison of non-dominated solutions by four algorithms. *Note: the red line denotes the theoretical Pareto front.* (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Detailed performance comparison of four algorithms.

	No. of FEs	No. of NSs	GD	HV	Computational time
NSGA-II	250	5	0.0099	0.1367	Less than 1 min
BOTR	250	25	0.0052	0.2810	Less than 1 min
BOEGO	250	46	0.0021	0.3081	About 4 h
BOSSO	250	54	0.0019	0.3140	Less than 1 min

Note: No. of FEs and No. of NSs denote the number of function evaluations and non-dominated solutions, respectively.

performed for each decision vector, while BOEGO employs the regressing Kriging model. The budgeted number of function evaluations is set to be 250.

Numerical results in Fig. 1 show that the non-dominated solutions by BOSSO are better-spread and closer to the theoretical Pareto front than those by NSGA-II and BOTR. BOSSO also exhibits a little better-spread performance than BOEGO, although their non-dominated solutions almost all locate on the theoretical Pareto front. Specifically, BOTR produces some solutions that are distant from the theoretical Pareto front, which implies that AAM cannot well smooth the stochastic noises. Even more, most solutions by NSGA-II are far away from the theoretical Pareto front, which means this type of algorithm (that relies on the extensive exploration of entire domain) is not applicable to solve the stochastic FON problem with limited function evaluations. More detailed comparison can refer to Table 1. That is, with fewer function evaluations and less computational time, BOSSO achieves more non-dominated solutions with a smaller GD and a higher HV.

Furthermore, Fig. 2 illustrates that BOSSO reaches the well convergence after about 110 function evaluations, and it always obtains the better-spread solutions that are closer to the theoretical Pareto front (i.e., a smaller GD and a higher HV) compared with NSGA-II and BOTR. Besides, although BOEGO also shows the well convergence, its corresponding GD and HV values are always a little larger and smaller than those by BOSSO, respectively. Thus, with limited computational costs, BOSSO is more applicable to address the bi-objective stochastic optimization problem than three others.

4.2. Real-field experiments

4.2.1. Construction of simulation model and its calibration

One urban road network in Changsha, China is modeled as the experimental scenario by the microscopic traffic simulator VISSIM. It includes 15 signalized intersections (with 47 signal phases), five non-signalized intersections, 29 entries, 24 exits, and 31 road links, c.f., Fig. 3. Before the network-wide signal timing optimization, it is the first step to calibrate the traffic network simulation model in the SO platform, which is built with VISSIM and Matlab through the COM interface. Here, traffic volume inputs of 29 entrances and three driving behavior-related parameters in the Wiedemann74 are set as the calibration parameters, i.e., the dimension of parameter vector $h=32$. Moreover, the upper and lower bounds of 29 flow inputs are determined according to the designed levels of road links, and those of three driving behavior parameters accord with the default upper and lower values in VISSIM. The benchmark data for the calibration cover the complete traffic vol-

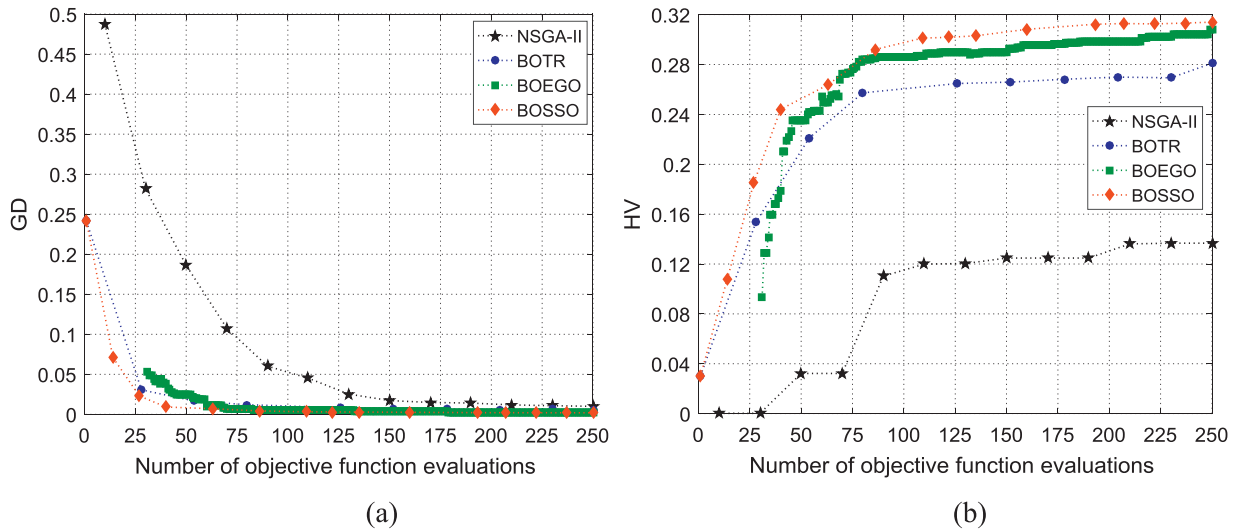


Fig. 2. Convergence performance comparison of four algorithms, (a) GD, (b) HV. Note: $R_H = [1.0, 1.0]$ for the calculation of HV.

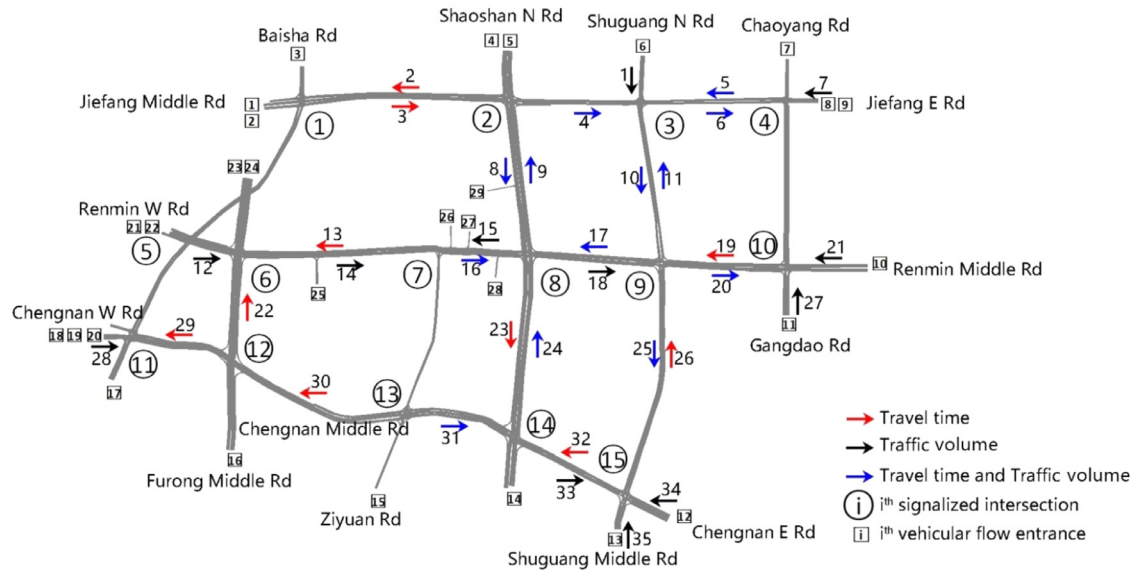


Fig. 3. VISSIM model of a regional network in Changsha, China.

umes in 25 links (i.e., $N_f = 25$), which are collected near the signalized intersections, and the complete travel time data in 23 links (i.e., $N_t = 23$) extracted from taxis' GPS. The two types of real-field data were collected during 9:00–9:30 a.m. on 2013/11/16 with time intervals of 5 and 2 min, respectively, which are then aggregated to denote the network-wide traffic states for this half an hour. The simulation time for a single run is about 90 s for the real traffic in half an hour.

Different from the single-objective calibration model in Zhang et al. (2017), the calibration here is to minimize the differences between the real-field and simulated traffic states from two aspects, which are evaluated by two Mean Absolute Percentage Errors (MAPEs) about traffic volume and travel time. Hence, the bi-objective stochastic calibration model is formulated as

$$\begin{aligned} \text{Minimize } f(p, \varsigma) &= \{E[MAPE_f(p, \varsigma)], E[MAPE_t(p, \varsigma)]\} \\ \text{s.t., } \Phi(p, \varsigma) &\rightarrow \{MAPE_f(p, \varsigma), MAPE_t(p, \varsigma)\} \end{aligned} \quad (14)$$

where $MAPE_f(p, \varsigma) = (\sum_{i=1}^{N_f} |F_i - FS_i(p, \varsigma)| / F_i) / N_f$ and $MAPE_t(p, \varsigma) = (\sum_{j=1}^{N_t} |T_j - TS_j(p, \varsigma)| / T_j) / N_t$ indicate the MAPE about traffic volume and travel time, respectively. p indicates the parameter vector to be calibrated. The search space is $\Omega_2 = \{p \in R^h: \mathbf{a}_2 \leq p \leq \mathbf{b}_2\}$, $\mathbf{a}_2$ and $\mathbf{b}_2$ are the lower and upper bounds of calibration parameters, respectively. F_i and $FS_i(p, \varsigma)$ indicate the field and simulated traffic volume in link i , respectively. T_j and $TS_j(p, \varsigma)$ denote the field and simulated travel time in link

Table 2
Calibrated parameters.

D1	D2	D3	V1	V2	V3	V4	V5
4.33	2.34	2.33	250	1356	112	2479	955
V6	V7	V8	V9	V10	V11	V12	V13
546	1805	395	124	1832	360	1373	2475
V14	V15	V16	V17	V18	V19	V20	V21
4209	148	801	471	137	214	1521	1851
V22	V23	V24	V25	V26	V27	V28	V29
2733	2788	2261	130	151	90	153	66

Note: D1-D3 denote three driving behavior-related parameters (D1 [m], D2-D3 [1]); V1-V29 [veh/h] respectively represent traffic volume inputs of 29 entrances in Fig. 3.

j , respectively. $\Phi(p, \varsigma)$ represents the microscopic traffic simulation with the inputs of p and ς . Besides, two other indices U_f and U_t (c.f., Eqs. (15,16)) are applied to judge the spatial distribution of calibration errors respectively about network-wide traffic flow and travel time, $U_f(U_t) \in [0,1]$. $U_f(U_t)=0$ indicates the perfect calibration, and $U_f(U_t)=1$ denotes that the field traffic states are not reproduced at all.

$$U_f(x, \varsigma) = \frac{\sqrt{\frac{1}{N_f} \sum_{i=1}^{N_f} (FS_i(x, \varsigma) - F_i)^2}}{\sqrt{\frac{1}{N_f} \sum_{i=1}^{N_f} (FS_i(x, \varsigma))^2 + \frac{1}{N_f} \sum_{i=1}^{N_f} (F_i)^2}} \quad (15)$$

$$U_t(x, \varsigma) = \frac{\sqrt{\frac{1}{N_t} \sum_{j=1}^{N_t} (TS_j(x, \varsigma) - T_j)^2}}{\sqrt{\frac{1}{N_t} \sum_{j=1}^{N_t} (TS_j(x, \varsigma))^2 + \frac{1}{N_t} \sum_{j=1}^{N_t} (T_j)^2}} \quad (16)$$

During the calibration experiment, $f_d=(0.2, 0.2)$ in $MAPE_f$ - $MAPE_t$ plane is set as the desired point of the decision maker in Appendix A. As we know, when the calibration errors about traffic volume and travel time reach about 20%, the calibrated microscopic traffic simulation model can well reproduce the reality. Other parameters of the BOSSO method are as follows: $d_{max}=80$, $d_{min}=40$, $\Delta^{(0)}=60$, $\Delta_{max}=100$, $\Delta_{tol}=20$, $\delta=0.01$, $\eta=10^{-8}$, $\tau_{dec}=0.6$, $\tau_{inc}=1.05$, $\mu=8.0 \times 10^{14}$, $\beta=6.0 \times 10^{14}$, $\omega=0.6$, $c_f=6.0 \times 10^{-5}$, and the budgeted simulation time is 1000. It takes approximately 24 h to run the BOSSO algorithm. Based on the numerical results, it is drawn that as for ultimate non-dominated solutions without filtering noises, the solution of ($MAPE_f=0.2406$, $MAPE_t=0.1876$) is the closest to f_d . As for ultimate non-dominated solutions after filtering noises, the solution of ($MAPE_f=0.2450$, $MAPE_t=0.2099$) is the nearest to f_d . These two solution points are satisfactorily and corresponding to the same calibration parameters, c.f., Table 2. After that, 300 repetitive simulations with these calibrated parameters are performed under stochastic environments, and their output bi-objective values (i.e., $MAPE_f$ and $MAPE_t$) are collected. Fig. 4(a,b) show that Gaussian density functions well fit the frequencies of $MAPE_f$ and $MAPE_t$, and two mean values (0.23097, 0.26206) are closer to the filtered solution point ($MAPE_f=0.2450$, $MAPE_t=0.2099$) than to the non-filtered solution point ($MAPE_f=0.2406$, $MAPE_t=0.1876$). This experimentally ensures the positive effect of the regressing Kriging model in mitigating noises from simulation outputs. Finally, the mean values of U_f and U_t , i.e., (0.11961, 0.13598), further validate the convincing and reasonable calibration results (c.f., Fig. 4(c,d)).

4.2.2. Network-wide signal timing optimization without coordination

Taking the calibrated network-wide traffic simulator as the evaluation tool, this section will optimize the network-wide signal timing plan without coordination, which is composed of green timings of 47 signal phases. The critical parameters of the BOSSO method are set as: $d_{max}=100$, $d_{min}=50$, $\Delta^{(0)}=77$, $\Delta_{max}=300$, $\Delta_{tol}=20$, $\delta=0.005$, $\eta=10^{-8}$, $\tau_{dec}=0.75$, $\tau_{inc}=1.2$, $\beta=6.0 \times 10^{16}$, $\omega=0.55$, $c_f=9.7 \times 10^{-5}$, and the budgeted number of simulations is 1000. What is noteworthy is that the non-interactive role of a decision maker is removed from the following experiments, which aims to diversify the non-dominated solutions. Meanwhile, three other algorithms (i.e., NSGA-II, BOTR, and BOEGO) are also implemented for the comparison. Experimental results in Fig. 5(a) show that these non-dominated solutions improve the initial bi-objective values ($T_r=1.0026$, $A_r=0.9312$) at various degrees, and only those by BOSSO all fall on the meta-Pareto front. It means that BOSSO is able to search more frontier non-dominated solutions. Meanwhile, GD* and HV by BOSSO respectively decreases and grows as the number of function evaluations increases, and become smaller and larger respectively than those by NSGA-II, BOTR and BOEGO in the latter iterations, c.f., Fig. 5(b,c). This implies the outperformance of BOSSO compared with these three counterparts under the same budgeted microscopic simulations.

In addition, the more detailed comparison can refer to Table 3. The fact that BOSSO costs more time than BOTR and NSGA-II to run 1000 microscopic simulations results from some extra computation time taken to build the regressing Kriging

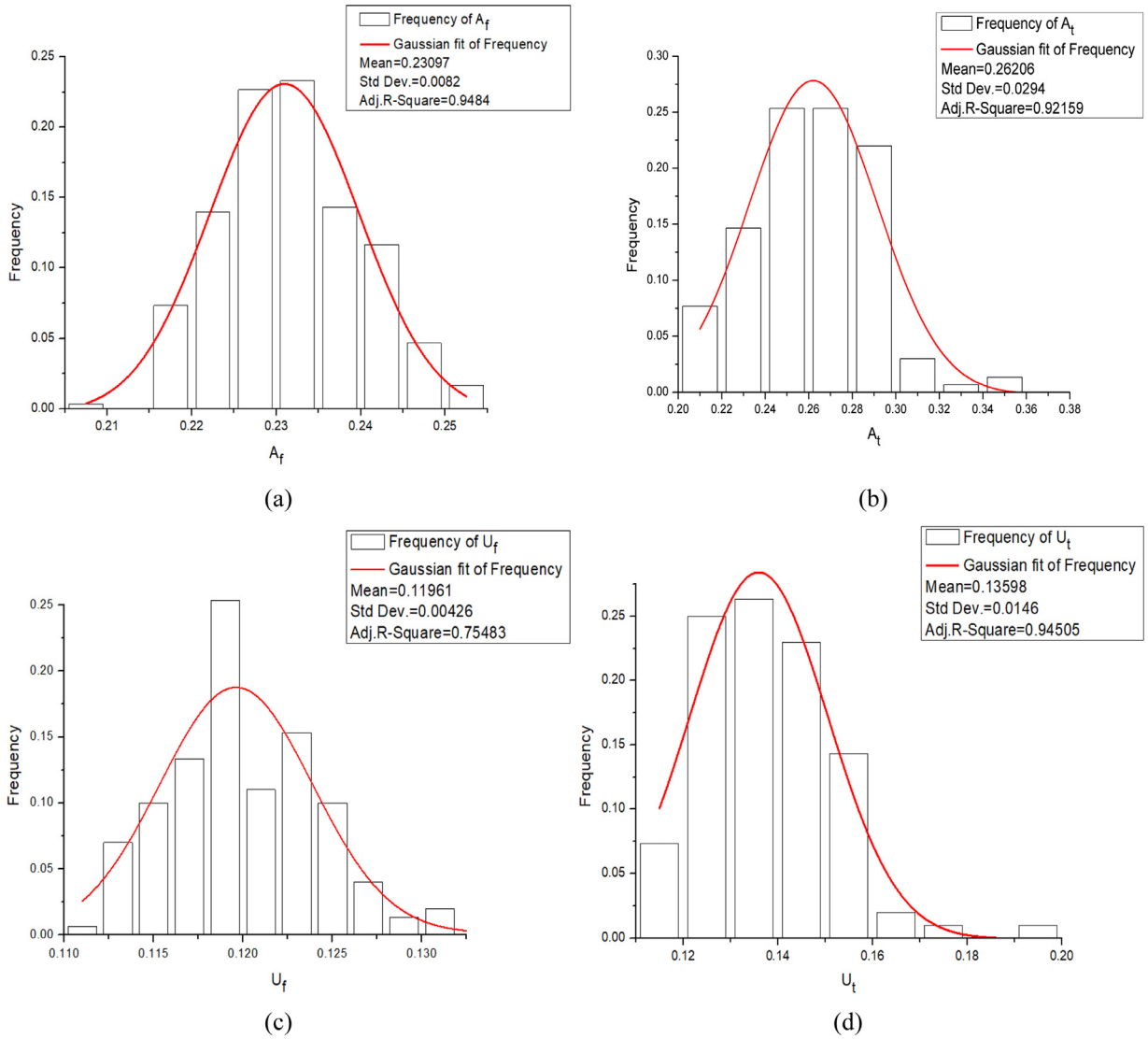


Fig. 4. Distributions of four measures, (a) A_f , (b) A_t , (c) U_f , (d) U_t .

Table 3

Detailed result comparisons of four algorithms for the non-coordinated case.

	No. of FEs	No. of NSs	GD*	HV	Computational time
NSGA-II	1000	3	0.0212	0.0934	22.9 h
BOTR	1000	7	0.0234	0.0919	21.2 h
BOEGO	1000	7	0.0142	0.1157	69.0 h
BOSSO	1000	7	0.0000	0.1243	28.2 h

model in BOSSO. However, the numerical results show that BOSSO performs better than BOTR and NSGA-II from both aspects of GD* and HV. That is also to say, BOSSO is able to search the comparative solutions as BOTR and NSGA-II do at the cost of the fewer microscopic simulations. When the microscopic simulation is costly enough, the time-saving value due to the saved simulations will surpass the extra time to construct the regressing Kriging model. In this condition, BOSSO can search the comparative (even better) non-dominated solutions as (than) BOTR and NSGA-II do with the fewer simulations and less computation time. Therefore, as long as BOSSO outperforms BOTR and NSGA-II under the same budgeted simulations, the superiority of BOSSO exists and will be enlarged when the microscopic simulation is sufficiently computationally expensive.

After that, the balance analyses are carried out for seven non-dominated solutions by BOSSO (c.f., Fig. 5(a)). Table 4 sorts these solutions in the increasing order concerning the relative average travel time T_r , and their signal timing plans are given

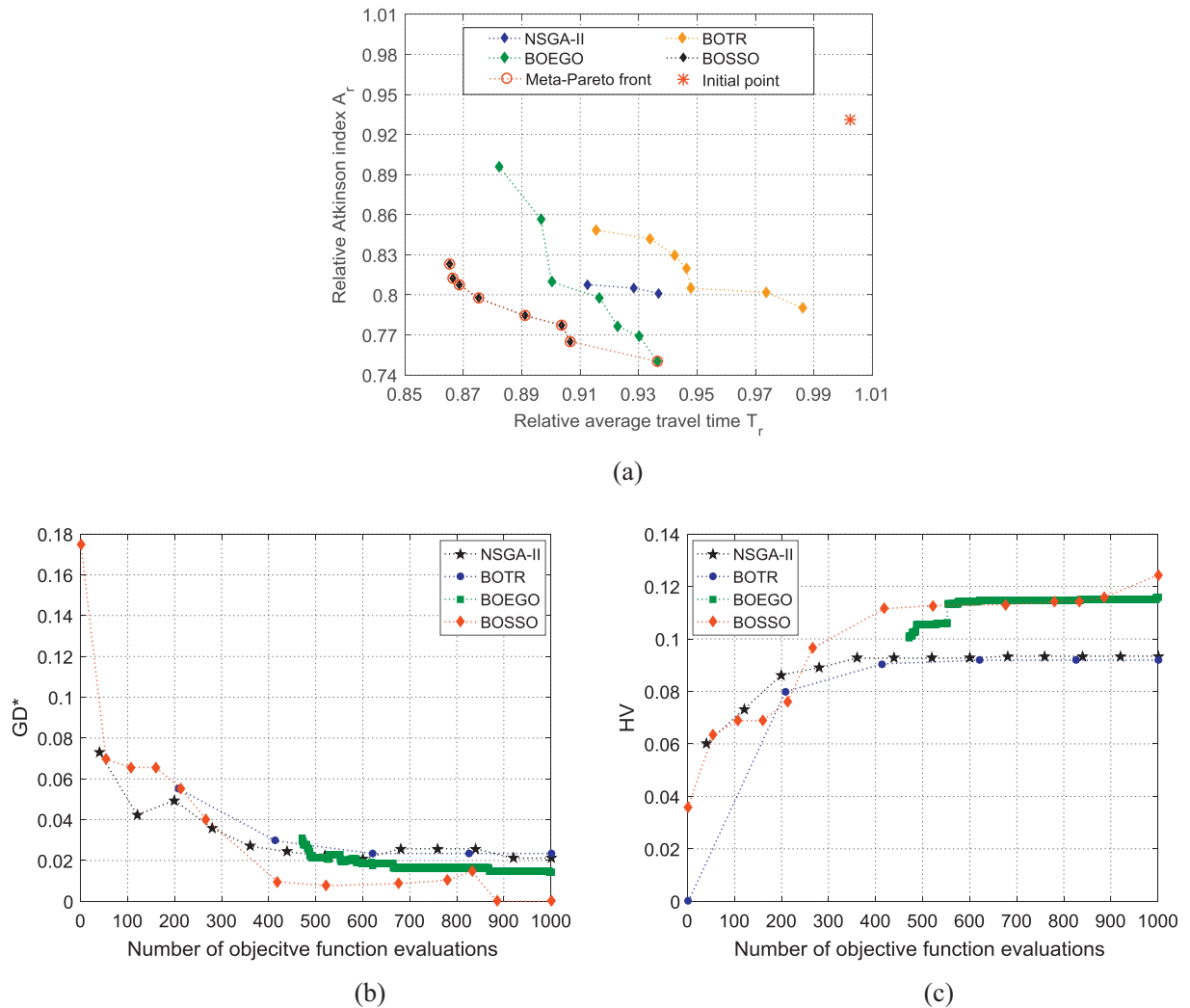


Fig. 5. Performance comparison of four algorithms for the non-coordinated case, (a) Non-dominated solutions, (b) GD^* , (c) HV. Note: $R_H = [1.1, 1.3]$ for the calculation of HV. Four sets of ultimate non-dominated solutions, GD^* and HV are based on solutions that dominate the initial point.

Table 4

Seven non-dominated solutions and their comparisons for non-coordinated case.

Non-dominated solution	T_r	A_r	(P_t, P_e) (%)	(P_t^+, P_e^+) (%)	(P_t^*, P_e^*) (%)
1	0.8652	0.8230	(0, 0)	(13.70↑, 11.61↑)	(13.48↑, 17.70↑)
2	0.8663	0.8120	(0.13↓, 1.34↑)	(13.59↑, 12.80↑)	(13.37↑, 18.80↑)
3	0.8687	0.8076	(0.40↓, 1.87↑)	(13.35↑, 13.27↑)	(13.13↑, 19.24↑)
4	0.8751	0.7974	(1.14↓, 3.12↑)	(12.71↑, 14.37↑)	(12.49↑, 20.26↑)
5	0.8911	0.7845	(2.99↓, 4.69↑)	(11.12↑, 15.76↑)	(10.89↑, 21.55↑)
6	0.9038	0.7773	(4.46↓, 5.56↑)	(9.85↑, 16.52↑)	(9.62↑, 22.27↑)
7	0.9065	0.7651	(4.77↓, 7.03↑)	(9.58↑, 17.83↑)	(9.35↑, 23.49↑)

Note: ↓ and ↑ respectively indicates the drop and improvement compared with the benchmark solution.

in Appendix C. Taking the 1st non-dominated solution as the benchmark, we obtain the drop (or improvement) degrees of the other six solutions with respect to T_r and A_r , which are denoted by the absolute percentage changes P_t and P_e , respectively. That is, when replacing the 1st solution with others, traffic efficiency is sacrificed to gain the improvement of traffic equity in a certain level. Meanwhile, P_t^+ and P_e^+ denote two absolute percentage changes respectively about T_r and A_r of these seven solutions compared with the initial point ($T_r = 1.0026$, $A_r = 0.9312$), and P_t^* and P_e^* represent those compared with the field bi-objective values ($T_r = 1.0$, $A_r = 1.0$). Obviously, the initial or field bi-objective traffic states are greatly improved by these seven solutions. Therefore, the bi-objectives can be simultaneously improved at various degrees, when the signal timing plan is not good enough, e.g., the field-implemented signal plan compared with the optimized

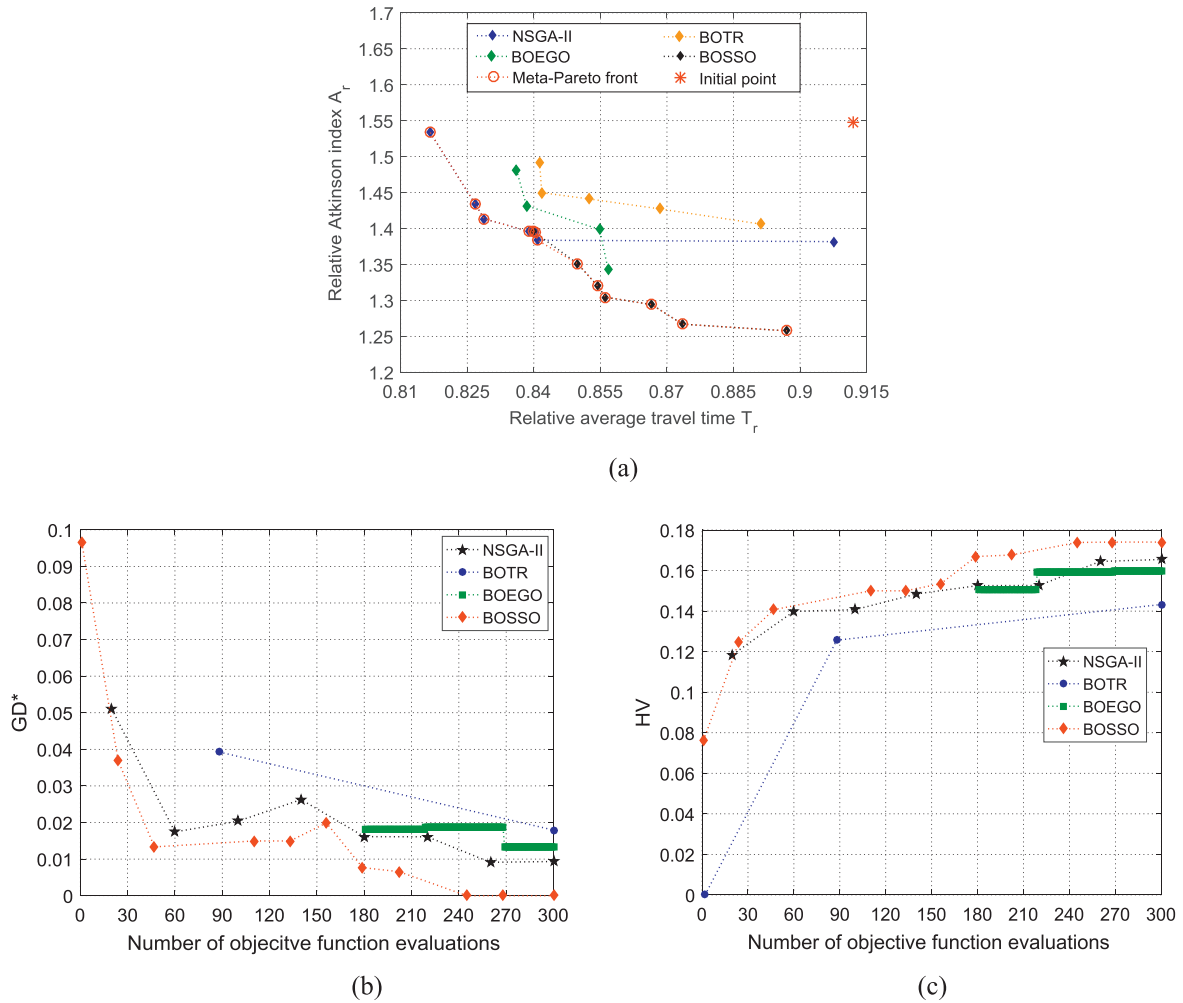


Fig. 6. Performance comparison of four algorithms for the coordinated case, (a) Non-dominated solutions, (b) GD^* , (c) HV. Note: $RH = [1.05, 2.10]$ for the calculation of HV. Four sets of ultimate non-dominated solutions, GD^* and HV are based on solutions that dominate the initial point.

ones. The competing relationship between bi-objectives also exists for these seven non-dominated solutions, that is, the improvement of traffic equity (traffic efficiency) sacrifices the traffic efficiency (traffic equity). In this condition, the preferred signal timing plan completely depends on the balance consideration of bi-objectives by the decision maker.

4.2.3. Network-wide signal timing optimization with coordination

Based on the signal timing results for the 5th non-dominated solution in Section 4.2.2 (c.f., Appendix C), this section will further optimize the coordinated signal plans for two arterial roads (i.e., *Furong Middle Rd* and *Shaoshan N Rd*) in the study site. So, the coordinated signal plan is composed of green timings of 15 signal phases and 3 offsets. Similarly, taking the calibrated traffic simulator as the evaluation tool, four algorithms (i.e., NSGA-II, BOTR, BOEGO and BOSSO) are used to solve the problem (6) with coordination. Notably, the critical parameters of BOSSO are set as: $d_{max}=40$, $d_{min}=20$, $\Delta^{(0)}=53$, $\Delta_{max}=100$, $\Delta_{tol}=10$, $\delta=0.005$, $\eta=10^{-8}$, $\tau_{dec}=0.75$, $\tau_{inc}=1.15$, $\mu=8.0 \times 10^{16}$, $\omega=0.6$, $c_f=1.4 \times 10^{-3}$, and the budgeted number of simulations is 300.

Numerical results in Fig. 6(a) demonstrate that ultimate non-dominated solutions by four algorithms improve the initial bi-objective values ($T_r=0.9120$, $A_r=1.5480$) at various degrees, and the meta-Pareto front is composed of all seven solutions by BOSSO and six solutions by NSGA-II. This validates the relatively better capability of BOSSO and NSGA-II in searching more frontier non-dominated solutions. Moreover, Fig. 6(b,c) illustrate that as the number of function evaluations increases, GD^* (HV) by BOSSO drops (increases) more rapidly than those by the other three algorithms, and the final GD^* (HV) by BOSSO is the smallest (largest). This successfully validates the better performance of BOSSO compared with the other three counterparts. Notably, GD^* and HV by BOSSO both have little variation once the number of function evaluations

Table 5

Detailed result comparisons of four algorithms for the coordinated case.

	No. of FEs	No. of NSs	GD*	HV	Computational time
NSGA-II	300	7	0.0093	0.1655	5.4 h
BOTR	300	5	0.0179	0.1433	5.0 h
BOEGO	300	4	0.0133	0.1599	11.3 h
BOSSO	300	7	0.0000	0.1741	5.9 h

Table 6

Seven non-dominated solutions and their comparisons for coordinated case.

Non-dominated solution	T_r	A_r	(P_{tc}, P_{ec}) (%)	(P_{tc}^+, P_{ec}^+) (%)	(P_{tc}^*, P_{ec}^*) (%)
1	0.8400	1.3956	(0, 0)	(7.89↑, 9.85↑)	(25.58↑, 7.84↓)
2	0.8499	1.3510	(1.18↓, 3.19↑)	(6.81↑, 12.73↑)	(24.71↑, 4.39↓)
3	0.8543	1.3207	(1.70↓, 5.37↑)	(6.32↑, 14.69↑)	(24.32↑, 2.05↓)
4	0.8562	1.3044	(1.93↓, 6.53↑)	(6.12↑, 15.74↑)	(24.15↑, 0.79↓)
5	0.8665	1.2942	(3.16↓, 7.27↑)	(4.98↑, 16.40↑)	(23.23↑, 0.00)
6	0.8735	1.2670	(3.99↓, 9.22↑)	(4.22↑, 18.16↑)	(22.61↑, 2.11↑)
7	0.8970	1.2576	(6.79↓, 9.89↑)	(1.64↑, 18.76↑)	(20.53↑, 2.83↑)

reaches about 240, which apparently shows the well convergence and the well enough solutions searched with such limited simulations. Thus, this study case further exhibits the superiority of BOSSO in finding the better non-dominated solutions.

Moreover, Table 5 gives the detailed result comparison of four algorithms. As for the reason why BOSSO costs a little more time than BOTR and NSGA-II to run 300 microscopic simulations can refer to the similar explanation in Section 4.2.2. In simple terms, as long as BOSSO performs better than BOTR and NSGA-II from both aspects of GD* and HV under the same budgeted simulations, the superiority of BOSSO exists and will be amplified when the microscopic simulation is computationally expensive enough.

Similar to Section 4.2.2, seven non-dominated solutions by BOSSO (c.f., Fig. 6(a)) are also sorted in the increasing order concerning the relative average travel time T_r (c.f., Table 6), and their signal timing plans are listed in Appendix D. By comparing the 1st non-dominated solution with the other six solutions, two absolute percentage changes P_{tc} and P_{ec} denote the drop (or improvement) degrees of T_r and A_r , respectively. Obviously, the improvement of traffic equity is paid off by the sacrifice of traffic efficiency at a certain degree. Thus, the competing relationship between bi-objectives exists, and how to choose the preferred coordinated signal plan depends on the balance consideration of bi-objectives by the decision maker.

Moreover, P_{tc}^+ and P_{ec}^+ denote two absolute percentage changes respectively about T_r and A_r of these seven solutions compared with the initial point ($T_r = 0.9120$, $A_r = 1.5480$), and P_{tc}^* and P_{ec}^* represent those compared with the field bi-objective values ($T_r = 1.0$, $A_r = 1.0$). Apparently, the seven non-dominated solutions significantly perform better than the initial point from both aspects of T_r and A_r , which validates the well optimization capability of BOSSO. Meanwhile, by comparing the 1st to 4th non-dominated solutions with the field bi-objective values, the significant improvement of traffic efficiency costs the traffic equity. However, as for the comparison of the 5th to 7th non-dominated solutions with the field bi-objective values, the optimized coordinated signal plans can significantly improve the traffic efficiency and meanwhile guarantee or even enhance the level of traffic equity.

4.2.4. Bi-objective comparison under variable traffic volumes

This section aims to compare the overall and local traffic efficiency and equity for three signal plans under variable traffic volumes. That is, the non-coordinated signal plan corresponding to the 5th non-dominated solution in Table 4, the coordinated signal plan corresponding to the 6th non-dominated solution in Table 6, and the field implemented signal plan. During numerical experiments, the traffic volume inputs of 29 entrances are set to fluctuate around their calibrated values in Table 2 with the variation ratio in the range of $[-20\%, 20\%]$. Then, from the lower bounds, traffic volume inputs of all entrances are simultaneously increased with the ratio of 2% for each step, in which the simulation experiments are performed for these three signal plans.

Numerical results in Fig. 7(a) illustrate that as the traffic volume grows, the average travel time for the whole road network (i.e., overall average travel time) gradually increases. The coordinated signal plan always produces the lower overall average travel time than the non-coordinated one, which is followed by the field implemented one. It is also clearly exhibited by Fig. 7(b), that is, the positive Relative Improvement Ratio (RIR) values by the coordinated signal plan are all higher than those by the non-coordinated one. Fig. 7(c) demonstrates that the Atkinson indices for the whole road network (i.e., overall Atkinson indices) for the real-field case and coordinated case have a growth trend with the increase of traffic volume, while those for the non-coordinated case only have a little fluctuation and are always smaller than those by the other two cases. Meanwhile, the coordinated signal plan usually performs worst. This is also clearly shown in Fig. 7(d). That is, the RIR values by the non-coordinated signal plan are all positive, but those by the coordinated signal plan are almost negative.

Thus, compared with the real-field case, the non-coordinated signal plan can simultaneously improve the overall traffic efficiency and overall traffic equity. Compared with the real-field case or non-coordinated case, the coordinated signal plan

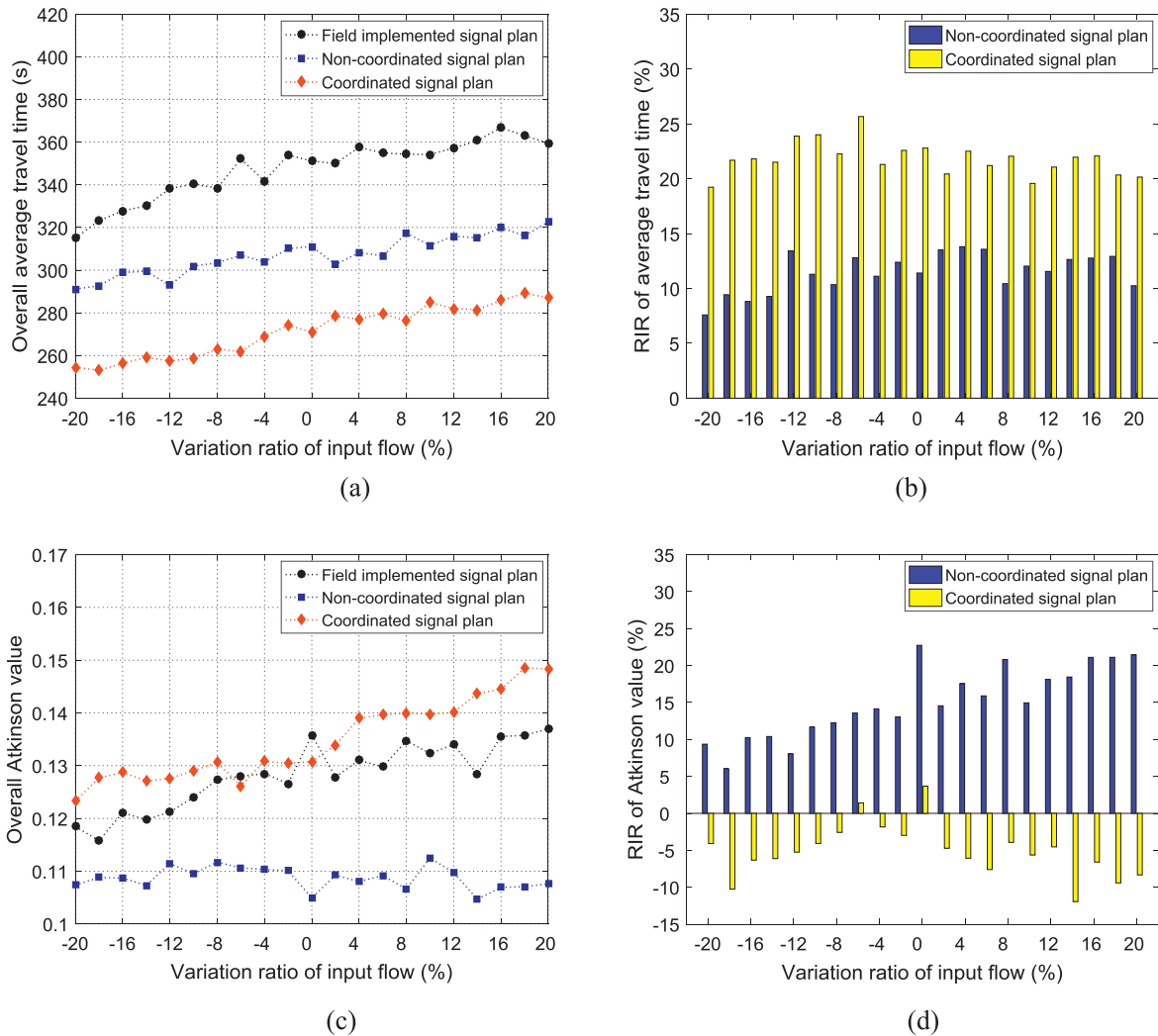


Fig. 7. Comparison of overall traffic efficiency and equity indices for three signal plans, (a) overall average travel time, (b) RIR of overall average travel time, (c) overall Atkinson index, (d) RIR of overall Atkinson index. Note: the positive (negative) RIR value indicates the better (worse) performance of the optimized signal plan compared with the field implemented signal plan, and the larger absolute value, the higher degree.

is capable to improve the overall traffic efficiency, but costs the overall traffic equity. This can be explained as follows. The coordinated signal plan added to two arterial roads benefits the right-of-ways of vehicles in the coordinated directions, but sacrifices those in other directions, which would worsen the overall Atkinson index. However, it is conducive to the improvement of overall traffic efficiency because of the priority provided for the higher traffic volumes in the coordinated directions of the two arterial roads.

Regarding two arterial signalized roads locally, it is not strange to find that the non-coordinated signal plan results in the higher local average travel time but mostly the lower local Atkinson index than the field implemented one (c.f., Fig. 8(a,c)). That is, the RIR values about the local average travel time are all negative, but those about the local Atkinson index are almost positive, c.f., the non-coordinated case in Fig. 8(b,d). This is because the model (6) treats the overall average travel time and overall Atkinson index as two optimization objectives. However, the coordinated signal plan can further reduce the local average travel time in almost all cases, but usually generate the higher local Atkinson index than the field implemented one and non-coordinated one (c.f., Fig. 8(a,c)). This is also clearly illustrated by Fig. 8(b,d). The reasons can also be briefly analyzed as follows. The coordinated signal plan applied for the north-south directions of *Furong Middle Rd* and *Shaoshan N Rd* sacrifices the right-of-ways of other approaches, which causes the worse local traffic equity. However, it helps to improve the local traffic efficiency due to the priority given to the higher traffic volumes in the coordinated directions of the two arterial roads.

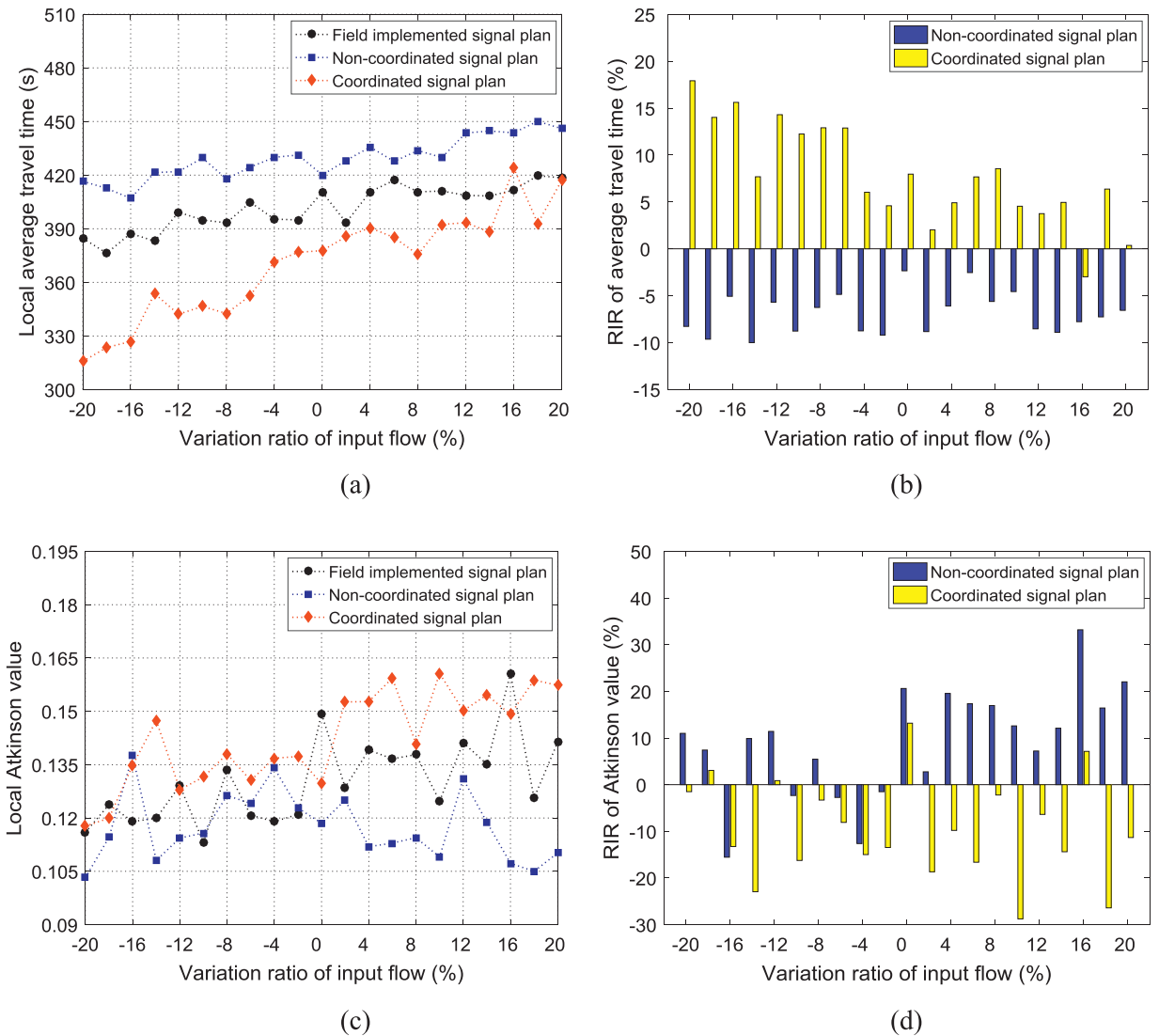


Fig. 8. Comparison of local traffic efficiency and equity indices for three signal plans, (a) local average travel time, (b) RIR of local average travel time, (c) local Atkinson index, (d) RIR of local Atkinson index.

In summary, under variable traffic volumes, the non-coordinated signal plan always performs better than the field implemented one in both traffic efficiency and equity from the overall aspect, but it is not that case from the local aspect. The coordinated signal plan usually improves the overall or local traffic efficiency compared with the field implemented and non-coordinated cases, but would cost the overall or local traffic equity. This also shows the existence of the competing relationship between bi-objectives.

5. Conclusions

The equity of right-of-way attracts more and more attentions of traffic management agencies, while the signal timing plan plays a direct impact on the traffic equity and efficiency of an urban road network. Meanwhile, the microscopic traffic simulator is able to provide the high-resolution evaluation and flexible experimental design. Therefore, this study proposes a network-wide signal timing stochastic SO model with the balance consideration of traffic equity and efficiency, which are respectively measured by the Atkinson index and average travel time. Then, the BOSSO method is put forward to solve this bi-objective stochastic SO problem, which include three critical features that deserve to be mentioned. That is: (1) the regressing Kriging model can tolerate the noises from the simulation outputs and well predict bi-objective values at a new sample. (2) An adaptive selector assigns a reasonable number of samples in the local trust-region to be evaluated by the microscopic simulation, which helps to well balance the computational costs and accuracies of quadratic regression models.

(3) Desired bi-objective values of a decision maker are set to guide the evolution of local trust-region, which contributes to generating more Pareto optimal solutions around the desired point. As for how to choose the desired bi-objective values, it completely depends on the specific bi-objective optimization problem and preference of the decision maker.

As for the benchmark problem of stochastic FON function, the BOSSO method is validated to perform better than three other algorithms (i.e., NSGA-II, BOTR and BOEGO) under the same budgeted number of simulations. In the real-field experiments, the traffic simulation model is firstly calibrated by BOSSO so as to reproduce the field network-wide traffic states indexed by travel time and traffic volume. Based on the well calibrated traffic simulator, the network-wide signal timing bi-objective stochastic SO problems with and without coordination are also solved by BOSSO. Numerical results indicate that compared with the field implemented case, the average travel time and Atkinson index for optimized non-coordinated signal plans are reduced by at most 13.48% and 23.49%, respectively, and those for optimized coordinated signal plans are reduced by at most 25.58% and 2.83%, respectively. It is also illustrated that under variable traffic volumes, the non-coordinated can satisfactorily reduce both average travel time and Atkinson index, and the coordinated one is able to further lower the average travel time but makes the Atkinson index become larger. The balance analyses show that the improvement of traffic equity (efficiency) costs the traffic efficiency (equity), i.e., the competing relationship between bi-objectives exists. Finally, the convergence performance indexed by GD* and HV illustrate that BOSSO outperforms NSGA-II, BOTR and BOEGO under the same budgeted simulations.

In summary, the proposed BOSSO method addresses well the network-wide bi-objective calibration and signal timing SO problems under uncertainties. It can also be easily transplanted to solve other similar problems. Future researches will focus on developing some efficient algorithms to cope with the SO problems with even more objectives in real world applications.

Acknowledgments

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Appendix A: Procedure of selecting the most isolated point

In the case that more Pareto optimal solutions are preferred around a desired point f_d , the procedure of selecting the most isolated point $x_c^{(k)}$ from the previous set of non-dominated solutions $\chi^{(k-1)}$ is designed as [Sub-Algorithm A1](#). The

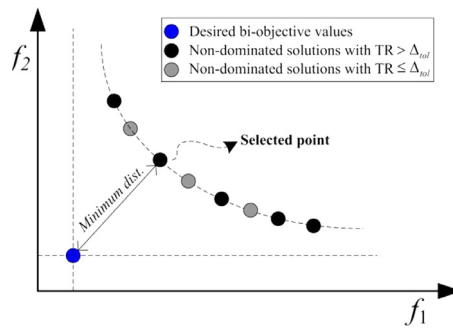


Fig. A1. The desired point dominates the previous set of non-dominated solutions. Note: TR denotes the trust-region.

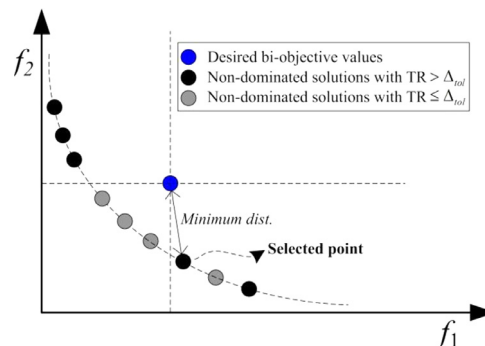


Fig. A2. Some non-dominated solutions dominate the desired point, but their trust-region radiuses are smaller than the tolerable value.

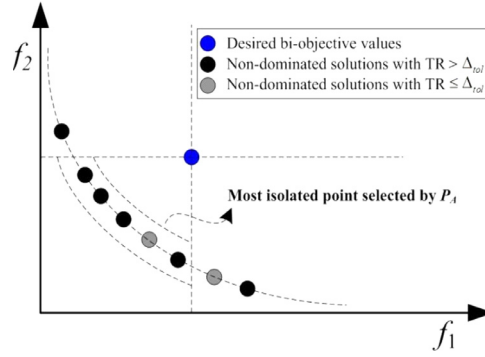


Fig. A3. Some non-dominated solutions dominate the desired point, and their trust-region radiuses are higher than the tolerable value. Note: P_A is the procedure of selecting the most isolated point by Ryu and Kim (2014).

Sub-algorithm A1

Select $x_c^{(k)}$ with a decision maker's non-interactive role.

INPUT

- 1: A decision maker's desired point f_d
- 2: The previous set of non-dominated solutions $\chi^{(k-1)}$
- 3: Set $\Theta = \emptyset$.

BEGIN

- 3: **For** each $x \in \chi^{(k-1)}$ and $\Delta^{(k-1)}(x) > \Delta_{tol}$ **do**
- 4: **If** $F(x) > f_d$ **then**
- 5: $\Theta \leftarrow \Theta \cup x$
- 6: **Else**
- 7: Θ is unchanged
- 8: **End if**
- 9: **End for**
- 10: **If** $\Theta = \emptyset$ **then**
- 11: $\gamma^{(k)}(x) = \|f_d - F(x)\|$, where $x \in \chi^{(k-1)}$ and $\Delta^{(k-1)}(x) > \Delta_{tol}$;
 $\gamma_{\min}^{(k)} = \min\{\gamma^{(k)}(x) | \Delta^{(k-1)}(x) > \Delta_{tol}, x \in \chi^{(k-1)}\}$;
 $\bar{x} = \arg \min\{\gamma^{(k)}(x) | \Delta^{(k-1)}(x) > \Delta_{tol}, x \in \chi^{(k-1)}\}$;
 $x_c^{(k)} = \bar{x}$ (c.f., Fig.A1 and Fig.A2).
- 12: **Else**
- 13: $\chi^{(k-1)} \leftarrow \Theta$
Implement P_A to select the most isolated point, c.f., Fig.A3.
- 14: **End if**
- OUTPUT**
- 15: The most isolated point $x_c^{(k)}$.

main idea is as follows. $x_c^{(k)}$ acts as the center of a trust-region, in which a poised set of samples will be generated, which may become Pareto optimal solutions as long as some conditions are satisfied. If f_d can guide $x_c^{(k)}$ to its neighborhood, more Pareto optimal solutions might exist around f_d , from which the decision maker can choose the most satisfactory solution. Thus, we have to firstly judge whether there are non-dominated solutions in $\chi^{(k-1)}$ with a higher trust-region radius than the tolerable value Δ_{tol} , which meanwhile dominate f_d . If exist, $x_c^{(k)}$ will be selected from such solutions by the selection procedure in Ryu and Kim (2014), c.f., Fig.A3. Otherwise, a non-dominated solution in $\chi^{(k-1)}$ that closest to f_d and with a higher trust-region radius than Δ_{tol} is selected as $x_c^{(k)}$ (c.f., Fig.A1 and Fig.A2). Notably, as for how to choose f_d , it completely depend on the preference of a decision maker, i.e., more consideration is given to the 1st objective or the 2nd objective, or equal consideration is given to each objective (Figs. A1–A3).

Appendix B: Regressing Kriging Model for Prediction and Re-estimation

To predict bi-objective values of untested samples or filter noises of bi-objective outputs from simulation, one classic surrogate model named regressing Kriging model is built based on global simulated samples. It owns a strong prediction capability and can smooth stochastic noises in simulation outputs. For bi-objective optimization problems, the main steps are given as follows.

- (1) **Assumptions:** To build the surrogate model for an expensive objective function $y_i = f_i(x)$, $x \in R^n$, where i indicates the i th objective and $i \in \{1, 2\}$, Gaussian stochastic process modeling is applied and the following assumptions are made

(Sacks et al., 1989; Jones et al., 1998; Rasmussen and Williams, 2006): For any x , $f_i(x)$ is a sample of following Gaussian random variable $\mu_i + \varepsilon_i(x)$, where $\varepsilon_i(x) \sim N(0, \sigma_i^2)$, μ_i and σ_i^2 are respectively constant mean and variance independent of x . That is, the prior distribution of $f_i(x)$ is $N(\mu_i, \sigma_i^2)$. For any $x, x' \in R^n$, the covariance $\text{Cov}[\varepsilon_i(x), \varepsilon_i(x')] = \sigma_i^2 c_i(x, x')$, where $c_i(x, x')$ is the Kriging basis function with the following correlation form

$$c_i(x, x') = \exp \left(- \sum_{k=1}^n \theta_{i,k} |x_k - x'_k|^{p_{i,k}} \right) \quad (\text{B1})$$

where $\theta_{i,k}$ indicates the importance of x_k in $f_i(x)$ ($\theta_{i,k} > 0$), and $p_{i,k}$ is related to the smoothness of x_k in $f_i(x)$ ($1 \leq p_{i,k} \leq 2$). For simplicity, $p_{i,k}$ is set as a constant value as 2.

- (1) **Hyperparameter Estimation:** For given K points $(x^1, x^2, \dots, x^K \in R^n)$ and their f_i -function values $(y_i^1, y_i^2, \dots, y_i^K)$, the hyperparameters μ_i, σ_i and $\theta_{i,k} (k = 1, 2, \dots, n)$ can be estimated by maximizing the following likelihood that $f_i(x) = y_i^l$ for $x = x^l (l = 1, 2, \dots, K)$:

$$\frac{1}{(2\pi\sigma_i^2)^{K/2} \sqrt{\det(C_i)}} \exp \left(- \frac{(y_i - \mu_i \mathbf{1})^T C_i^{-1} (y_i - \mu_i \mathbf{1})}{2\sigma_i^2} \right) \quad (\text{B2})$$

where C_i is $K \times K$ matrix whose (l_1, l_2) -element is $c_i(x^{l_1}, x^{l_2})$, $y_i = (y_i^1, y_i^2, \dots, y_i^K)^T$ and $\mathbf{1}$ is a K -dimensional column vector of ones. To maximize Eq. (B2), the estimated values of μ_i and σ_i^2 should respectively be

$$\hat{\mu}_i = \frac{\mathbf{1}^T C_i^{-1} y_i}{\mathbf{1}^T C_i^{-1} \mathbf{1}} \text{ and } \hat{\sigma}_i^2 = \frac{(y_i - \mathbf{1} \hat{\mu}_i)^T C_i^{-1} (y_i - \mathbf{1} \hat{\mu}_i)}{K} \quad (\text{B3})$$

By substituting Eq. (B3) into Eq. (B2), the unknown parameters μ_i and σ_i are eliminated from Eq. (B2). As a result, the likelihood function depends only on $\theta_{i,k} (k = 1, 2, \dots, n)$.

- (1) **The best linear unbiased prediction and predictive distribution:** For given $\hat{\theta}_{i,k}$, $\hat{\mu}_i$ and $\hat{\sigma}_i^2$, a regularization constant λ_i is added to the diagonal of the correlation matrix C_i so as to regress (tolerate) noisy observations under uncertainties (He et al., 2017), that is, the new correlation matrix is $\tilde{C}_i = C_i + \lambda_i I$, where I is an identity matrix. With the regressing Kriging model, the predicted mean value and estimation variance at a new point x are respectively given by

$$\hat{f}_i^r(x) = \hat{\mu}_i^r + r_i^T \tilde{C}_i^{-1} (y_i - \mathbf{1} \hat{\mu}_i^r) \quad (\text{B4})$$

$$s_{i,r}^2(x) = \hat{\sigma}_{i,r}^2 \left[1 + -r_i^T \tilde{C}_i^{-1} r_i + \frac{1 - \mathbf{1}^T \tilde{C}_i^{-1} r_i}{\mathbf{1}^T \tilde{C}_i^{-1} r_i} \right] \quad (\text{B5})$$

where $\hat{\mu}_i^r = \frac{\mathbf{1}^T \tilde{C}_i^{-1} y_i}{\mathbf{1}^T \tilde{C}_i^{-1} \mathbf{1}}$, $\hat{\sigma}_{i,r}^2 = \frac{(y_i - \mathbf{1} \hat{\mu}_i^r)^T \tilde{C}_i^{-1} (y_i - \mathbf{1} \hat{\mu}_i^r)}{K}$ and $r_i = (c_i(x, x^1), c_i(x, x^2), \dots, c_i(x, x^K))^T$.

Also, $N(\hat{f}_i^r(x), s_{i,r}^2(x))$ can be regarded as a predictive distribution of $f_i(x)$ for f_i -function value y_i^l at x^l for $l = 1, 2, \dots, K$. Ultimately, the regularization constant λ_i can be optimized simultaneously with the scaling coefficient parameter $\theta_{i,k}$ by maximizing the likelihood function Eq. (B2) with the Matlab function *fmincon*.

Based on the built regressing Kriging model, Eq. (B4) is used here to re-estimate bi-objective values of simulated samples and to predict bi-objective values of unevaluated samples in the local trust-region. In this way, the noises from simulation outputs can be mitigated, and a lot computational costs can be saved when a large number of samples are required to fit three meta-models with high dimensions (c.f., Eq. (9)).

Appendix C: Network-wide signal plans for seven non-dominated solutions in Table 4 are given in Table C1

Table C1.

Table C1

Seven optimized signal plans without coordination.

Note: $SI\ i$ indicates the i th signalized intersection in Fig.3; $ST\ i$ denotes the signal timing plan corresponding to the i th non-dominated solution in Table 4; SP represents the signal phases.

	$SI\ 1$		$SI\ 2$				$SI\ 3$		$SI\ 4$	
SP										
$ST1$	21	41	35	16	56	55	76	61	102	55
$ST2$	22	45	30	13	53	60	72	63	106	54
$ST3$	23	45	27	13	47	57	72	64	110	55
$ST4$	25	47	29	10	50	60	75	66	108	57
$ST5$	28	42	33	10	51	46	64	61	102	52
$ST6$	31	42	34	16	54	59	73	60	102	51
$ST7$	30	42	34	15	45	54	70	61	103	52

	$SI\ 5$			$SI\ 6$					$SI\ 7$			$SI\ 8$			
SP															
$ST1$	59	34	45	60	11	31	26	44	43	74	40	47	42	35	53
$ST2$	57	35	47	67	13	39	23	47	52	80	49	44	44	37	54
$ST3$	58	35	47	62	12	33	23	47	46	77	49	44	44	43	54
$ST4$	60	38	50	65	10	36	20	50	49	80	46	47	47	40	57
$ST5$	49	30	42	57	21	33	27	36	45	70	55	43	42	35	52
$ST6$	55	33	45	60	13	30	26	44	50	74	41	41	41	34	52
$ST7$	55	41	45	70	15	31	25	48	54	75	51	41	42	35	52

	$SI\ 9$				$SI\ 10$			$SI\ 11$		$SI\ 12$		
SP												
$ST1$	42	72	45	19	29	35	52	64	24	51	40	26
$ST2$	39	74	47	15	38	38	53	67	23	55	44	33
$ST3$	39	74	47	17	37	37	52	69	23	59	43	31
$ST4$	42	77	50	14	35	40	55	70	20	56	46	30
$ST5$	44	68	42	18	26	32	48	61	39	48	42	25
$ST6$	47	77	45	19	37	38	49	65	26	53	45	25
$ST7$	47	72	45	10	30	35	50	65	25	51	41	35

	$SI\ 13$		$SI\ 14$				$SI\ 15$			
SP										
$ST1$	57	26	50	20	50	15	16	41	64	20
$ST2$	53	33	41	22	53	13	13	44	67	18
$ST3$	53	29	41	23	58	13	13	49	67	17
$ST4$	56	31	44	25	56	10	10	46	70	15
$ST5$	55	41	34	22	52	14	12	37	55	24
$ST6$	51	25	39	28	51	15	12	40	64	21
$ST7$	51	30	39	20	51	16	15	41	65	20

Appendix D: Network-wide signal plans for seven non-dominated solutions in Table 6 are listed in Table D1

Table D1.

Table D1

Seven optimized signal plans with coordination.

Note: SI 2, SI 8 and SI 14 belong to Shaoshan N Rd and their common cycle length is 184 s. SI 6 and SI 12 belong to Furong Middle Rd and their common cycle length is 183 s. The coordinated direction is the north-south direction, and thus SI 2 (SI 6) is set as the benchmark intersection for coordinating Shaoshan N Rd (Furong Middle Rd). Signal plans for other signalized intersections accord with ST 5 in Table C1.

SI 2						SI 8						SI 14					
SP					Offset					Offset					Offset		
ST1	16	16	45	95	--	44	38	31	59	42	33	43	24	72	91		
ST2	10	24	54	84	--	54	26	24	68	45	45	51	12	64	98		
ST3	27	15	48	82	--	49	14	43	66	43	49	27	32	64	87		
ST4	27	13	53	79	--	51	17	49	55	37	51	36	33	52	89		
ST5	25	17	46	84	--	47	27	40	58	31	47	40	30	55	86		
ST6	27	13	50	82	--	51	17	43	61	35	51	34	39	48	95		
ST7	30	10	54	78	--	52	23	54	43	37	53	33	52	34	92		

SI 6						SI 12				
SP						Offset				Offset
ST1	17	22	19	21	95	--	73	35	66	28
ST2	21	29	10	29	85	--	82	24	68	35
ST3	20	23	15	35	81	--	82	36	56	32
ST4	17	27	11	37	82	--	79	39	56	34
ST5	22	29	16	34	73	--	77	39	58	30
ST6	15	29	13	38	79	--	79	38	57	34
ST7	15	27	12	39	81	--	80	35	59	45

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